SMK budi luhur
TEKNOLOGI INFORMASI, SENI DAN INDUSTRI KREATIF

PENGAJUAN PEMBIAYAAN LOMBA MAGE 12 - ITS

A. Jenis Lomba: IoT COMPETITION

B. Latar Belakang

Tim kami telah berpartisipasi dalam kompetisi MAGE 12 dan berhasil lolos pada tahap proposal. Saat ini, kami telah memasuki tahap semi final di mana kami akan melaksanakan Tahap 2: Pengumpulan Karya (Final Works & Portfolio). Batas waktu (deadline) untuk pengumpulan Tahap 2 adalah pada 19 Oktober 2026, pukul 23.59. Dokumentasi dukungan terkait pencapaian ini dapat dilihat pada lampiran.

C. Peserta

NO	Nama	Keterangan
1	Muhammad Zahir Siraj	Tim Trilogy
2	Zabdan Harjati Purnomo	Tim Trilogy
3	Muhammad Rasya Al Faridzi	Tim Trilogy

D. Biaya

No	Nama Barang	Deskripsi Singkat (Penjelasan Alat)	Link	Qty	Total Biaya Alat
1	Biaya Pendaftaran Mage 12	Biaya pendaftaran kompetisi	-	1	Rp95,000
2	ESP32-S3 Microcontroller	Otak utama (mikrokontroler)	https://tk.tokopedia.com/ZSVLYR5wt/	1	Rp120,000
3	Expansion Board ESP32	Papan penghubung	https://tk.tokopedia.com/ZSVLFCvKm/	1	Rp120,000
4	Capacitive Soil Moisture V2	Sensor untuk mendeteksi tingkat kelembapan	https://tk.tokopedia.com/ZSVLFT8NQ/	1	Rp15,000
5	Soil Moisture Sensor Cover	Casing pelindung	https://tk.tokopedia.com/ZSVLY6Xjv/	1	Rp17,000
6	Sensor DHT22 (AM2302)	Sensor untuk mengukur suhu	https://tk.tokopedia.com/ZSVXAuCGJ/	1	Rp25,000
7	Sensor MPU-6050 (GY-521)	Sensor untuk mendeteksi kemiringan	https://tk.tokopedia.com/ZSVLY5Feu/	1	Rp40,000
8	Modul Transceiver LoRa	Modul komunikasi nirkabel	https://tk.tokopedia.com/ZSVLYg8kC/	2	Rp150,000
9	Baterai Li-ion 18650	Baterai isi ulang	https://tk.tokopedia.com/ZSVLYP46N/	2	Rp70,000
10	Cover / Holder Baterai 18650	Tempat Baterai	https://tk.tokopedia.com/ZSVjogwPV/	1	Rp12,000
11	Solar Panel Mini (5V/6V)	Panel surya	https://tk.tokopedia.com/ZSVLYvQTh/	2	Rp50,000
12	Led COB Fleksible 3V	Lampu emergency	https://tk.tokopedia.com/ZSVX1eaNM/	2	Rp15,000
13	Relay 1 Channel 3V	Saklar elektrik pengatur daya	https://tk.tokopedia.com/ZSVX115E3/	1	Rp20,000
14	Kabel Jumper Female to Female	Kabel penghubung	https://tk.tokopedia.com/ZSVxnc1Ns/	1	Rp10,200

15	Kabel Jumper Female to Male	Kabel penghubung	https://tk.tokopedia.com/ZSVxnc1Ns/	1	Rp10,200
16	Kabel Jumper Male to Male	Kabel penghubung	https://tk.tokopedia.com/ZSVxnc1Ns/	1	Rp10,200
17	Modul Charger Baterai TP4056	pengisian daya baterai Li-ion	https://tk.tokopedia.com/ZSVCGbXoa/	1	Rp5,890
18	Breadboard	Papan uji coba	https://tk.tokopedia.com/ZSVxn3HLp/	1	Rp7,800
19	Junction box ip 65 200x120	Box untuk menampung semua alat	https://tk.tokopedia.com/ZSVcSxbSV/	1	Rp72,500
20	Buzzer Speaker Active 3V	Speaker	https://tk.tokopedia.com/ZSVXA8nPw/	1	Rp1,900
21	Alumnium Profile	Penyangga alat	https://tk.tokopedia.com/ZSVXDe6U4/	50CM	Rp50,000
22	KY-026 Flame Sensor Module	Pendeteksi api	https://tk.tokopedia.com/ZSVcdRnjs/	1	Rp7,000
23	SMA FEMALE, KABEL 20CM	Kabel Antena	https://tk.tokopedia.com/ZSVcdgTp7/	2	Rp25,000
24	KABEL SERABUT AWG 24	Kabel penghubung	https://tk.tokopedia.com/ZSVcR6VUq/	1	Rp4,200
25	Voltage DC 0-25v	Check Persentase baterai	https://tk.tokopedia.com/ZSVcjppW6/	1	Rp9,000
26	Cable Gland PG7	cable waterproof	https://tk.tokopedia.com/ZSVcrwnSc/	10	Rp11,500
27	Bracket Corner	Penyangga alat	https://tk.tokopedia.com/ZSVchEhsX/	5	Rp10,000
28	T NUT M5	Baut	https://tk.tokopedia.com/ZSVckrpmF/	5	Rp7,000
29	Baut L M5	Baut	https://tk.tokopedia.com/ZSVckU2ej/	15	Rp9,000
30	Mur Nilon M5	Baut	https://tk.tokopedia.com/ZSVck9J3h/	15	Rp8,000
31	Lem Silikon	Lem	https://tk.tokopedia.com/ZSVcBNXCd/	1	Rp25,000
32	tongkat antena 433mhz	Antena	https://tk.tokopedia.com/ZSVcBVJe9/	2	Rp110,000
33	RING KARET M5	Baut	https://tk.tokopedia.com/ZSVcS8aT9/	10	Rp10,000
	Subtotal				Rp1,153,390
	Dana Darurat (20%)				Rp230,678
	Total Keseluruhan				Rp1,384,068

Tangerang, 31 Agustus 2026

Hormat kami,

Pembimbing	Pengaju (Ketua Tim)
Ferdyudiman, M.Kom	**Muhammad Zahir Siraj**

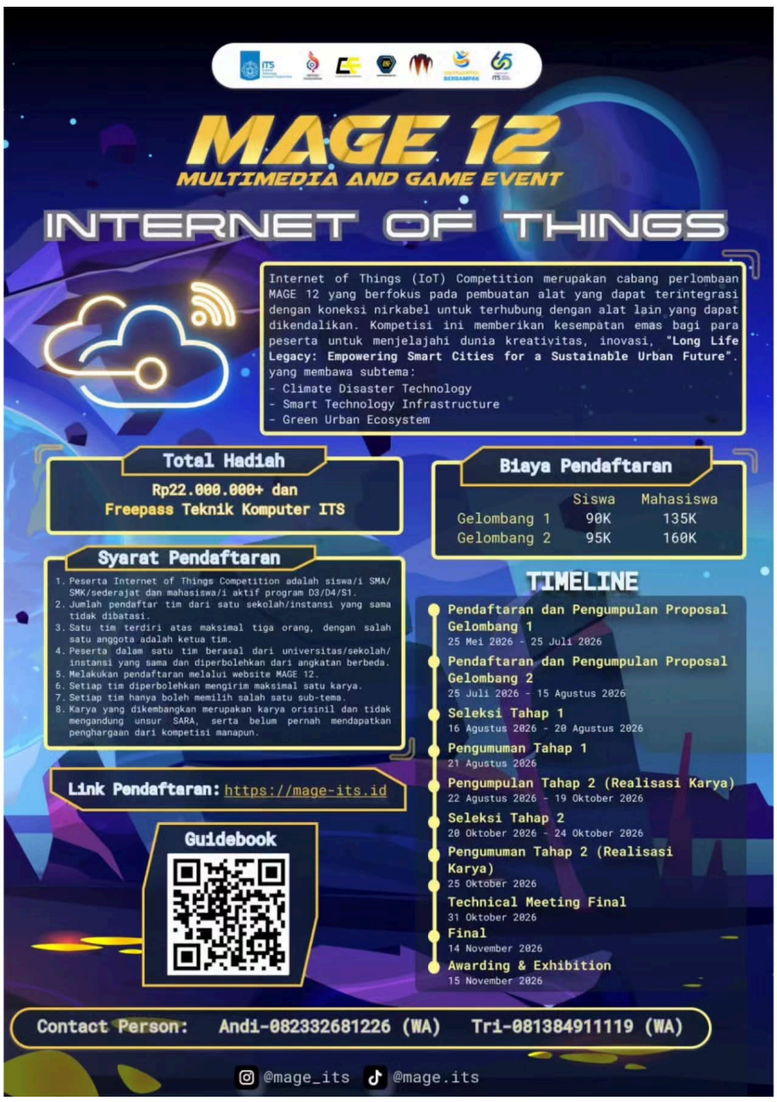
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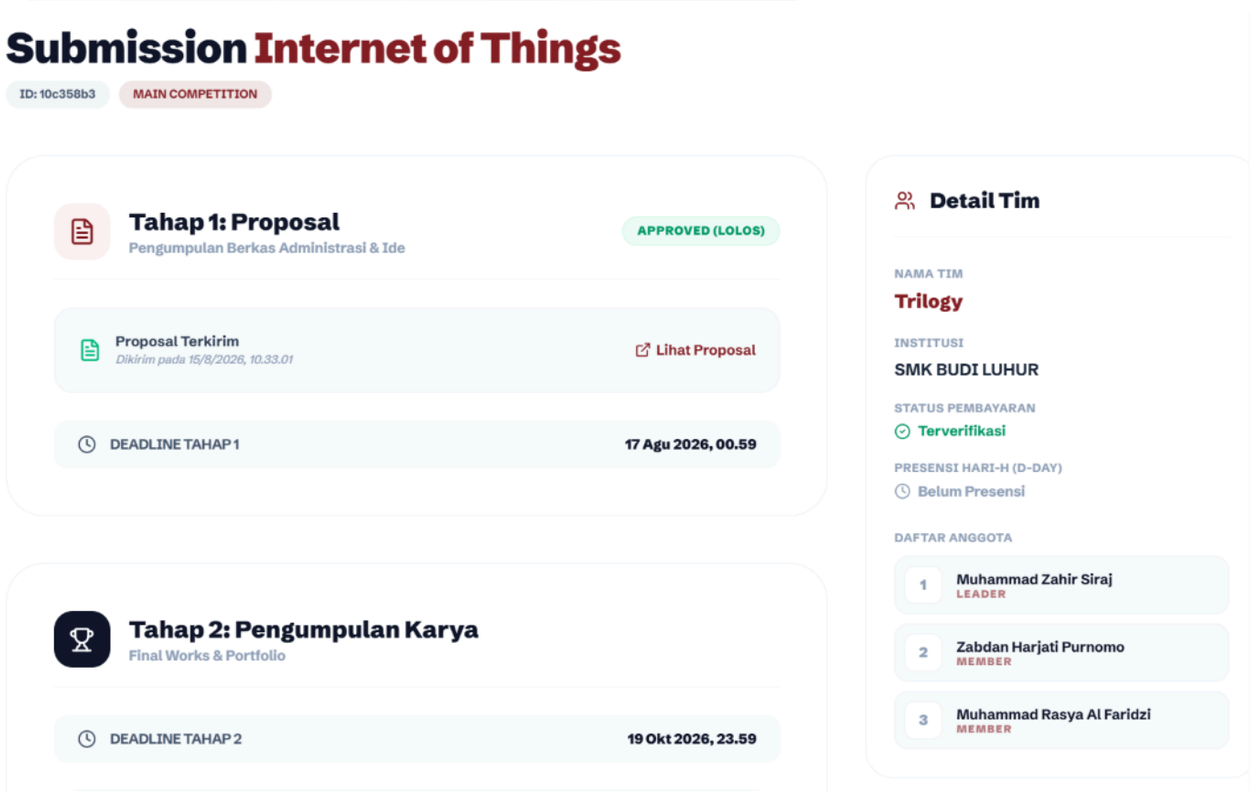
Kepala Sekolah	Waka Bidang Kesiswaan
Joko Waluyo,S.Si.,M.Pd.	**Denok Susanti,S.S**

Menyetujui,

Direktur Akademik	Direktur Operasional
Anastasia Eliarti Estari, M.	**Miming Roeswidiradjatin**

Lampiran 1. Pendukung Proposal





Lampiran 2. Biodata Tim Lomba

1. Biodata Anggota Tim

Nama Lengkap	Muhammad Zahir Siraj
Jurusan	Rekayasa Perangkat Lunak
NIS	240127041
Tempat, Tanggal Lahir	Mojokerto, 31 December 2008
Kegiatan yang Diikuti	1. Anggota SLC Div. Civic 2. Ketua Kelas RPL (X) 3. Wakil Ketua Kelas RPL (XI, XII) 4. Wakil Ketua P5 (Learn it and Educate it) 5. World GreenMech Contest 6. Robot Regional Competition 7. Robot National Competition 8. Robotik TIK-INFORMATIKA NASIONAL 9. IoT POLINES 10. CREW EO APPMBGI PT. TRICATHA SEMPITERNAL ASIA 11. Panitia Technopreneur 7.0 (Merch)
Penghargaan	1. Silver Winner Robot Regional Competition 2. Bronze Winner Robot National Competition 3. Excellent Performance World GreenMech Contest 4. Juara II Robotik TIK-INFORMATIKA NASIONAL 5. Lulusan Terbaik Angkatan 44 SMP PGRI 2 6. Penerima Beasiswa Budi Luhur 24/25 7. Juara 2 Design IoT 8. Juara 3 IoT

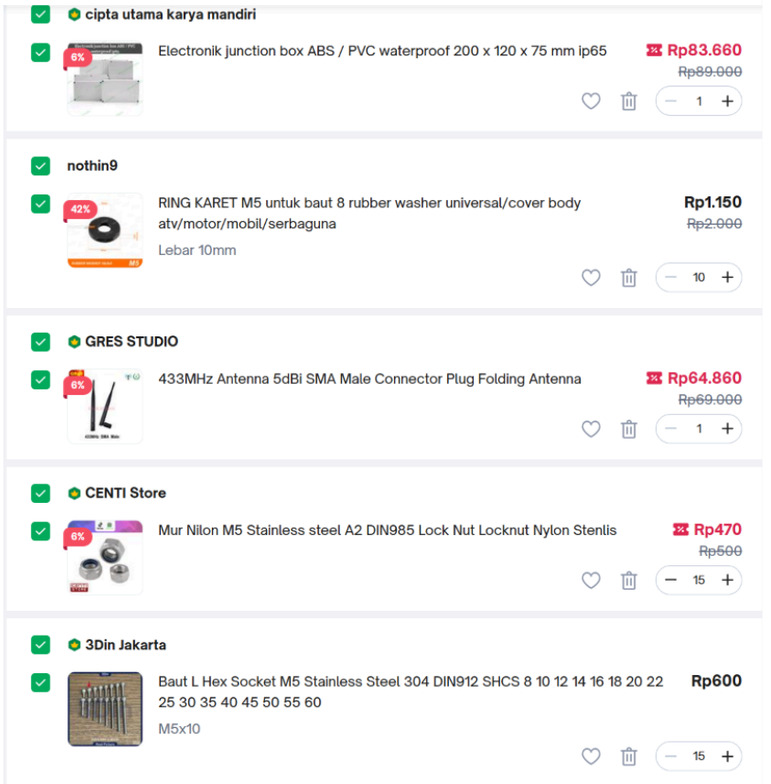
2. Biodata Anggota Tim

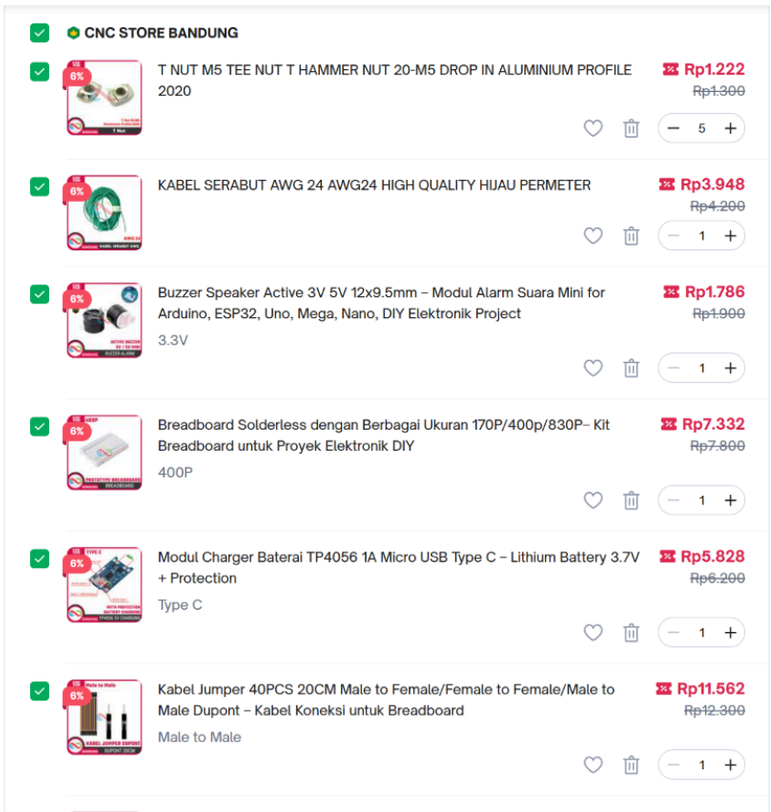
Nama Lengkap	Muhammad Rasya Al Faridzi
Jurusan	Rekayasa Perangkat Lunak
NIS	240127038
Tempat, Tanggal Lahir	Bogor, 24 Juli 2009
Kegiatan yang pernah Diikuti	1. Anggota SLC Div. Broadcast 2. Panitia Bluvocation (Sekretaris) 3. Panitia Technopreneur 7.0 (Koor Lomba) 4. Lomba KSR 2024 cabang Geografi (Finalis) 5. Guest Star Tosca Pensi 2026 6. Guest Star Technopreneur 6.0 7. Stem Competition 2023 8. Festival Olimpiade Nasional 9. Alseace 2025 cabang Band 10. Isaac Newton Comp. cabang Fisika
Penghargaan	1. Festival Olimpiade Nasional cabang Fisika (Top 10 Banten) 2. Isaac Newton Comp. cabang Fisika (Gold Medal) 3. Alseace 2025 cabang Band (Top 5) 4. Kompetisi Sains Ruang Guru cabang Geografi (Finalis/Top 200)

3. Biodata Anggota Tim

Nama Lengkap	Zabdan Harjati Purnomo
Jurusan	Rekayasa Perangkat Lunak
NIS	240127068
Tempat, Tanggal Lahir	Jakarta, 11 Agustus 2008
Kegiatan yang Diikuti	1. MAGE 11 ITS Game Development 2. Lomba Roket Air Kabupaten Magelang
Penghargaan	1. Juara 3 Harapan Lomba Roket Air Kabupaten Magelang

Lampiran 3. Gambar Alat







Kabel Jumper 40PCS 20CM Male to Female/Female to Female/Male to Male Dupont – Kabel Koneksi untuk Breadboard

Male to Female

Rp9.588
~~Rp10.200~~

1



Kabel Jumper 40PCS 20CM Male to Female/Female to Female/Male to Male Dupont – Kabel Koneksi untuk Breadboard

Female to Female

Rp9.588
~~Rp10.200~~

1



Modul LoRa RA-02 SX1278 433MHz SPI + Antena – Modul Wireless Jarak Jauh untuk Arduino & IoT

+ Antenna

Rp82.344
~~Rp87.600~~

1



MPU6050 MPU-6050 3 AXIS ANALOG GYROSCOPE ACCELEROMETER MODULE

Rp51.606
~~Rp54.900~~

1



DHT22 AM2302 Sensor Suhu & Kelembaban – Temperature Humidity Module untuk Arduino, ESP32 & IoT DIY

Rp22.184
~~Rp23.600~~

1



Capacitive Soil Sensor Tanah Hygrometer Humidity Moisture Module V1.2 / V2.0 – Sensor Kelembaban Tanah untuk Proyek DIY

V2.0

Rp13.442
~~Rp14.300~~

1



Sisa 5

Expansion Base Plate ESP32-S3 – GPIO Expansion Board / Papan Ekspansi Pin untuk Proyek IoT

Rp126.900
~~Rp135.000~~

~~Rp135.000~~

1



ESP32-S3 DevKitC-1 N16R8 Type-C dengan PSRAM 16MB – Board Pengembangan untuk IoT

Rp125.866
~~Rp133.900~~

1

jocky jaya



20x20Mm Alumunium Bracket Corner Breckets 2020

Rp1.786
~~Rp1.900~~

5

Glodok123



Cable Gland / Kabel Gland PG7 3.5~6mm PG9 PG11 PG13.5 PG16 PG19 PG21

PG7, Putih

Rp1.081
~~Rp1.150~~

10

solarperfect



Sensor Tegangan Voltage DC 0-25v Pendeteksi Volt Measurement Module

Rp8.178
~~Rp8.700~~

1

PLATINUM Audiophile



Kabel Pigtail MHF4 to RP SMA RPSMA Male Female Konektor PM-11 PM-12 PM-13 PM-14

Rp12.500



2. SMA FEMALE, KABEL 20CM

2

art techno new_NEW

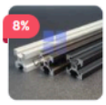


KY-026 Flame Sensor Module

Rp6.486
~~Rp6.900~~

1

Vlinder Aluminium



V Slot 2020 Alumunium Extrusion Profile - Silver Anodized

Silver, 10 cm

Rp8.084
~~Rp8.800~~

5

TOKO RYTECH



Relay 1 Channel 3V 3.3V Relay Board Modul 1CH

Rp20.000

1

Beeze Part



Led COB Fleksible 3V Panjang White Putih 6000k Flexible 40 titik mata

Rp7.500

4

Buzper Official



Solar Panel Mini 5V 9V 12V Tenaga Surya Matahari Power Solar Cell DIY Poly Crystalline

5V - 1.1W

Rp20.492
~~Rp21.800~~

Buzper Official



Solar Panel Mini 5V 9V 12V Tenaga Surya Matahari Power Solar Cell DIY Poly Crystalline

5V - 1.1W

Rp20.492
~~Rp21.800~~

2

DH TRONIC



Box Holder Battery Batree Batre 18650 x 2 Slot PARALLEL / SERI Tempat Baterai + Tutup Saklar On-Off

PARALLEL

Rp17.500

1

Rakki Panda Indonesia



RakkiPanda Baterai 18650 Button Top & Flat Top 3000mAh

Button Top

Rp55.000

2

De-Micro



Capacitive Soil Moisture Sensor v2.0 Cover

Rp15.300

1